



RE24 – Advanced Battery Technology

Battery Technology: Provides a comprehensive experimentation system for battery technology.

With the equipment students will fully understand which battery technologies is required and how this decision is made based on capacity, loading and durability.

The kit is supplied with different battery technologies such as lead, NiMH, or Lithium-Polymer (LiPo) as well as a PEM-fuel cell.

PRODUCT CODE: RE24

Categories: Educational, RE Series - Renewable Energy, RE24 Battery Technology

Tags: Battery , Battery Technology , Capacitor (Super Cap) , Fuel Cell , Lead , LiPo , Lithium-Polymer(LiPo) , NiMH , Pb , PEM-Fuel Cell Charger , RE24 , Renewable Energy , Sustainable Energy

DESCRIPTION

The RE24 Advanced Battery Technology system provides students with a self-contained modular system, covering the physical fundamentals of storage technology necessary for the transition to sustainable energy.

The system allows the characteristics of different battery types to be analysed along with their relevant suitable application.

The system is supplied with different battery technologies such as lead, NiMH and Lithium-Polymer(LiPo) as well as a PEM-Fuel Cell Charger Module and for the correct determination of internal resistance four-terminal sensing is possible.

TECHNICAL SPECIFICATIONS

- 1 x Motor module Pro
- 1 x Battery module NiMH 3xAAA Pro
- 1 x Capacitor module Pro
- 1 x Base unit Professional
- 1 x Resistor module (triple) Pro
- 1 x Resistor plug element 1 Ohm

- 1 x Resistor plug element 100 Ohm
- 3 x Resistor plug element 10 Ohm
- 1 x Resistor plug element 33 Ohm
- 1 x Lithium-polymer (LiPo) battery module
- 1 x Battery module holder 1xAAA Pro
- 1 x Battery adapter cable
- 1 x Fuel cell holder Pro
- 1 x Lead (Pb)-battery module Pro
- 1 x Electric model car
- 1 x LiFePo-battery AAA
- 1 x Charger module
- 1 x AV-module
- 1 x Potentiometer module 110 Ohm Pro
- 1 x Distilled water (100ml)
- 1 x Propeller
- 1 x Safety test lead, 50cm, red
- 1 x Safety test lead, 50cm, black
- 1 x Safety test lead, 25cm, red
- 1 x Safety test lead, 25cm, black
- 1 x NiZn-battery AAA
- 3 x Safety short-circuit plug, with mid socket
- 1 x Digital multimeter
- 1 x Reversible fuel cell
- 1 x Aluminium case
- 1 x NiMH battery AAA

FEATURES & BENEFITS

- Training system for storing renewable energy
- Battery types: **NiMH**, **NiZn**, **Pb**, **LiFePo**, **LiPo**, **capacitor** (super cap) and **fuel cell**
- All components are prepared for four terminal sensing
- Including charger module
- Modular design
- Supplied in a self-contained aluminium case
- Includes in-depth manual and predefined experiments

Experimental Content

- Setup of a simple circuit
- Ohm's law
- Series connection of ohmic resistances
- Parallel connection of ohmic resistances
- Start-up and idling behaviour of a motor
- Nominal voltage and capacity of voltage sources
- Four-terminal sensing
- Internal resistance of voltage sources
- Series connection of voltage sources
- The capacitance of a battery module



- The energy density of battery modules
- The Ri efficiency of a battery module
- The total efficiency of a battery module
- Temperature-dependent behaviour of the lithium-polymer cell
- The charging process of a capacitor
- The discharge process of a capacitor
- I-V characteristics of the single NiMH battery module
- I-V characteristics of the NiZn battery module
- I-V characteristics of the LiFePo battery module
- I-V characteristics of the lead battery module
- I-V characteristics of the lithium-polymer battery module
- I-V characteristics of the triple NiMH battery module
- The charging process of the NiMH battery
- The charging process of the NiZn battery
- The charging process of the LiFePo battery
- The charging process of the lead battery
- The charging process of the lithium-polymer battery
- The discharging process of a battery module
- Hydrogen production in the reversible hydrogen fuel cell
- Characteristic curve of the electrolyzer
- Hydrogen consumption of a fuel cell
- Characteristic curve of the fuel cell
- The efficiency of the hydrogen fuel cell
- Operation of the electric car with several battery modules
- Operation of the electric car with the reversible fuel cell

Related Curriculums

- Renewable Energies
- Electrical Engineering
- Automotive Engineering

Other products in the series

RE10: Advanced Photovoltaic Energy

RE12: Advanced Wind Energy

RE14: Advanced Fuel Cell Technology

RE16: Advanced Thermal Energy

RE18: Advanced Smart Grid Technology

Operational Conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz



- Level and stable work surface

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 7Kg

Dimensions

TRAY

- **Length:** 0.640m
- **Width:** 0.165m
- **Height:** 0.370m

Order Codes

RE24: Advanced Battery Technology

YOU MAY ALSO LIKE...



**RE22 –
BIOENERGY**

[Read more](#)



**RE12 –
ADVANCED
WIND ENERGY**

[Read more](#)



**RE14 –
ADVANCED
FUEL CELL
TECHNOLOGY**

[Read more](#)



RE20 – BIOFUEL

[Read more](#)



**RE16 –
ADVANCED
THERMAL
ENERGY**

[Read more](#)



**RE18 –
ADVANCED
SMART GRID
TECHNOLOGY**

[Read more](#)

Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781